

README — Task 14: Fairness, Bias Audit & Explainability

Objective

- Measure outcome disparities across relevant groups and establish a defensible, continuously monitored fairness framework.

Analysis Performed

- Group-level outcome analysis
- Outcome-rate comparison
- Absolute disparity measurement
- Group-rate ratio measurement
- Sample-size analysis
- Explainability coverage
- Human-review coverage
- Continuous fairness monitoring

Methodology

- Validate the dataset, define relevant groups and outcomes, calculate group-level rates, compare observed differences, document metric choices and limitations, and establish a monitoring process. Observed disparities describe differences in outcomes but do not independently establish their cause.

Key Metrics

- Outcome rate is calculated for each group. Absolute disparity is expressed in percentage points. A rate ratio compares the minimum observed group rate with the maximum observed group rate. Sample sizes and data limitations are considered before interpretation.

Output

- Group-level outcome tables, disparity measures, monitoring metrics, visualizations and a final report.

Tools

- Python, Pandas, NumPy, Matplotlib, ReportLab and Google Colab.

Data Limitation

- The analysis uses actual uploaded records when available. No synthetic production observations are created. Results depend on the quality and completeness of the supplied data.

Conclusion

- The project provides a repeatable framework for measuring outcome disparities and monitoring fairness continuously, with findings suitable for hand-off to AI/ML and Compliance teams.